5.2

Maximum Likelihood Estimators

Idea

A maximum likelihood estimator (MLE) estimates an unknown parameter with the value whose associated model is most likely to have produced the observed data.

The joint density

In general, for fixed $\vec{x} = (x_1, x_2, \ldots, x_n)$, the joint density is defined as $f(\vec{x}; \theta) = f(x_1, x_2, \ldots, x_n; \theta)$.
Note that θ is assumed fixed in the density function since its value is conditional on θ.

Joint density versus likelihood function

Mathematically, the joint density and the likelihood function are identical. Conceptually, the likelihood function differs from the joint density in that θ is assumed to be unknown and the data, $\vec{x} = (x_1, x_2, \ldots, x_n)$ are fixed.

The likelihood function

The likelihood function is denoted
$L(\theta) = L(\theta; \vec{x}) = f(\vec{x}; \theta)$.
Note that θ is assumed fixed in the density function since its value is conditional on θ.

Definition 5.2.1

Suppose that $X_1, X_2, \ldots, X_j n$ have joint pdf $f(\vec{x}; \theta)$.
A **likelihood function** is denoted by $L(\theta)$ is defined as any function proportional to $f(\vec{x}; \theta)$ when thought of as a function of θ.
The **maximum likelihood estimator** of θ is the value of θ that maximizes $L(\theta)$.

5.2.1

Examples

Example 5.2.2

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} \text{Bernoulli}(p)$. Determine the MLE of p.

Example 5.2.2 (cont)

Example 5.2.2 (cont)

Notation

Given a likelihood function $L(\theta)$, we denote the log-likelihood function by $\ell(\theta) = \ln L(\theta)$.
The maximum likelihood estimator for θ that maximizes $L(\theta)$, and equivalently, $\ell(\theta)$, since $\ln$ is an order-preserving transformation.

Example 5.2.3

Let $X_1, X_2, \ldots, X_n$ be a random sample with pdf
$f(\vec{x}; \theta) = \theta x^{\theta - 1} I_{(0,\infty)(x)}, \quad \text{for } \theta > 0.$
Determine the MLE of θ.

Example 5.2.3 (cont)

Example 5.2.3 (cont)

Example 5.2.4

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$. Determine the MLEs for μ and σ^2.

Example 5.2.4 (cont)

Example 5.2.4 (cont)

Example 5.2.4 (cont)

Example 5.2.4 (cont)

Example 5.2.5 (Parameters in the Support of the Distribution)

Let $X_1, X_2, \ldots, X_n$ be a random sample from the distribution with pdf
$f(\vec{x}; \theta, \lambda) = \theta \lambda^{\theta} x^{-(\theta+1)} I_{(\lambda, \infty)}(x)$,
with $\theta > 0, \lambda > 0$. Find the MLEs for θ and λ.

Example 5.2.5 (cont)

Example 5.2.5 (cont)

Example 5.2.5 (cont)

Example 5.2.5 (cont)

Example 5.2.5 (cont)

5.2.2

The Invariance Property of MLEs

Definition 5.2.2

If θ is the MLE of θ, then the MLE of a function $\tau(\theta)$ is
$\widehat{\tau(\theta)} = \tau(\hat{\theta})$.
This is known as the **invariance property** of MLEs.

Invariance examples

Invariance examples (cont)

Psuedo proof of invariance

Psuedo proof of invariance (cont)